AITJ44-COIMBATORE

CLASS:XI TOPIC:TUPLE

SUB: COMPUTER SCIENCE[083] Fac.Code: **IRN**

Q1. What do you mean by Tuple in Python?

Q2. Write a statement to create an empty tuple named 'T1'.

Q3. Write a statement to create a tuple 'T1' containing first five even numbers.

Q4. Write the code to convert the given list L1 to tuple. L1 = [1, 2, 3, 4, 5]

Q5. Write a statement to create tuple T1 with single element.

Q6. Write the output of the following code :

```
T1 = (4)
print(type(T1))
```
<class 'int'>

Q7. Write the output of the following code :

```
T1 = 4, 5, 6
print(type(T1))
```
<class 'tuple'>

Q8. Write the output of the following code :

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1)
print(T1 * 2)
print(T1 + T1)
print(len(T1) * 2)
```
(1, 2, 3, 4, 5, 6, 7, 8)
(1, 2, 3, 4, 5, 6, 7, 8, 1, 2, 3, 4, 5, 6, 7, 8)
(1, 2, 3, 4, 5, 6, 7, 8, 1, 2, 3, 4, 5, 6, 7, 8)
16

Q9. Write a statement to create tuple T1 with the following data.

1, 4, 'CS', 'IP', 5

T1 = (1, 4, 'CS', 'IP', 5)

Q10. Write the output of the following code :

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[0])
print(T1[-1])
print(T1[2 + 3])
print(T1[4 - 1])
print(T1[7%2])
```
1
8
6
4
2

print (tup)

. Write the output of the following :

```
1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[ : ])
print(T1[3 : ])
print(T1[ : 4])
print(T1[-2 : -5 ])
```

(1, 2, 3, 4, 5, 6, 7, 8)
(4, 5, 6, 7, 8)
(1, 2, 3, 4)
( )

Q12. Write the output of the following :

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[1 : : 2])
print(T1[-1 : -5 : -2])
print(T1[: : -1])
print(T1[ : 7 : 2])
```

(2, 4, 6, 8)
(8, 6)
(8, 7, 6, 5, 4, 3, 2, 1)
(1, 3, 5, 7)

Q13. Which error is returned by the following code :

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[15])
```

Index error : tuple out of range

Q14. Write two differences between Tuple and List.

Q15. Write a statement in python to concatenate the given tuples.

T1 = (1, 2, 3)

T2 = (5, 6, 7)

T1 + T2

Q16. Write the output of the following code :

```
T1 = (45, 67, 98)
T1 = T1 + (1, 2, 3)
print(T1)
```

(45, 67, 98, 1, 2, 3)

Q17. Write the output of the following code :

```
T1 = (45, 67, 98)
T1 = T1 * 3
print(T1)
```

(45, 67, 98, 45, 67, 98, 45, 67, 98)

Q18. Write the output of the following code :

```
T1 = (45, 67, 98)
T2 = ((45, 67, 98))
print(T1 in T2)
print(45 in T2)
print(45 in T1)
print(T1 + T2)
```

False
True
True
(45, 67, 98, 45, 67, 98)

print (tup)

Q19. Explain the following functions in reference to Tuple in Python.

1. len( )

2. count( )

Q20. Write a program to accept five numbers from the user and store it in a tuple 'T1'.

Q21. Write a program to accept five fruit name from the user and store it in a tuple 'F1'.

Q22. Write a program to store 'n' number of sports in a tuple 'S1'. (Accept 'n' from the user)

Q23. Consider the following tuple and write the code for the following statements :

T1 = (12, 3, 45, 'Hockey', 'Anil', ('a', 'b'))

a. Display the first element of 'T1'
b. Display the last element of 'T1'
c. Display 'T1' in reverse order.
d. Display 'Anil' from tuple 'T1'
e. Display 'b' from tuple 'T1'

Q24. Write the output of the following :

```
for i in tuple("SHIMLA"):
    print(i + i)
```

Q25. Write the output of the following :

```
>>> 7 in tuple("1234567890")
>>> '7' in tuple("1234567890")
```

Q26. Write the output of the following :

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(len(T1) + T1[-1])
print(T1[T1.count(23) + len(T1) -5])
print(T1.count(T1[6]))
print(T1.count(max(T1)))
```

Q27. Write the output of the following :

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(max(T1))
print(min(T1))
```

Q28. Write the output of the following :

```
T1 = ('Hockey', 'Cricket', 'Football')
print(max(T1))
print(min(T1))
```

Write the output of the following :

```
T1 = ('Raman', 'Ram', 'Ramaiya')
print(max(T1))
print(min(T1))
```

Q30. Write the output of the following :

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(sorted(T1))
print(sorted(T1[2 : 7]))
print(T1.index(23))
print(T1.index(23, 3, 9))
```

Q31. Write the output of the following :

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(sum(max(T1) + min(T1)))
```

Q32. Write a program to accept three numbers from the user and insert it at the end of given Tuple T1.

T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)

Q33. What type of error is returned by following statement :

```
>>> (a,b,c,d) = (5, 6, 8)
```

Q34. Write the output of the following code :

```
>>>(name, rollNo, subject)  = ("Anil", 9, "CS")
>>> name
```

Q35.Predict the output:

```
T=(1,2,3)

del T

print(T)
```

Q36. What do you mean by Nested Tuple? Give one example

Q37. Write a statement to print 30 from the given tuple.

```
a = ("Seventy", [1, 2, 3], (20, 30, 40), "Eighty")
```


print (tup)

Q38. Write a statement to unpack the following tuple into 3 variables.

(60, 70, 80) (n1, n2, n3) = (60, 70, 80)

Q39. Write a program to swap the values of given tuples.

T1 = ("A", "B") T1 = ("A", "B") print (T1)

T2 = (34, 65) T2 = (34, 65) print (T2)

T1, T2 = T2, T1

Q40. Write the output of the following code :

```
T1 = ("A",["B","D"],"C")
T1[1][1] = "C"
print(T1)
```

("A", ["B", "C"], "C")

Q41. Write a program to print the frequency of a number accepted from the user in given tuple :
T1 = (12, 17, 18, 25, 19, 12, 18, 5)

Q42. Write a program in python to concatenate all the characters of given tuple. T1 = ('B' , 'O', 'O', 'K')

Expected OUTPUT : BOOK

Q43. Write a program to remove a number (accepted from the user) from the given tuple T1 = (12, 15, 18, 21, 24, 27, 30)

SAMPLE EXECUTION

Enter element to remove : 21
Tuple after removing element is : (12, 15, 18, 24, 27, 30)

Q44. What do you mean by Unpacking Tuple? Give example

Q45. What do you mean by Packing Tuple? Give Example

Q46. Write one similarity between Tuples and String.

Q47. Write the output of the following :

```
T1 = (2, 'Apple', 6)
print(max(T1))
```

Type error

Q48. Consider the given tuple and write the output of given statements: T1 = (1, 23, 4, 5, "A", ["C", "D"], (23, 45), 45)

1. **print(len(T1))** 8
2. **print(T1.index(45))** 7
3. **print(T1.count(45))** 1
4. **print(T1[5][0]*2)** CC
5. **print(T1[:5])** (1, 23, 4, 5, 'A')
6. **print(T1[5:])** (["C", "D"], (23, 45), 45)
7. **print(T1.index("A"))** 4

8. **print(T1[-1 : -7 : -2])**
9. **print(T1[4] + T1[5])**
10. **print(max(T1))**

Q49. Write the length of following tuple :

T1 = (1,(2,3),(((34,56),45,67),39)) 3

Q50. Write a program to store the detail (Roll number, Name, Section) of three students in a tuple. Accept all data from the user.

SAMPLE OUTPUT :
((1, 'Amita', 'B'), (2, 'Sunita', 'C'), (3, 'Daya', 'A'))

Q51. Write a program to accept roll number from user and display it's detail from given tuple

Det=((1, 'Amita', 'B'), (2, 'Sunita', 'C'), (3, 'Daya', 'A'))

Q52. Write a program to accept five numbers from the user and store these numbers in a tuple. Display the following from tuple :

1. **Largest number**
2. **Smallest number**
3. **Sum of all numbers**
4. **Average of all numbers**

Q53. Write a program in python to print the second largest element in tuple given below :

T1 = (23, 45, 87, 45, 67, 43, 23, 12)

Q54. Write a program in python to display the sum of all even numbers and odd numbers separately from the tuple given below :

T1 = (24, 45, 87, 46, 67, 44, 23, 12)

Q55. Write the output of the following :

T1 = ("Amit", "Sumit")
T2 = "Sumit", "Amit"
T3 = "Amit", "Sumit"
print(T1==T2) False
print(T1==T3) True

Q56. Write a program to accept a number from the user and store first 10 multiples of number in a tuple for example :

SAMPLE OUTPUT :

Enter any number : 4
Tuple = (4, 8, 12, 16, 20, 24, 28, 32, 36, 40)

Q57. Write a program to accept a string from the user and convert it into tuple. for example

SAMPLE OUTPUT :

Enter any String : tuple
Tuple = ('t', 'u', 'p', 'l', 'e')

print (tup)

Q58. Write a program to display the sum of all the numbers in a given tuple :

T1 = (1, 2, 3, 'A', 'B', 4, 'Sun')

Q59. Write a program to find the sum of marks of all students stored in the given tuple.

T1 = (("Suman", 75),("Glory" , 35),("Ravi",50))

Expected Output is : 160

Q60. Write the output of the following :

T1 = (1, (2, 'a'), (3, 'b', 'c'), (7, 'd', 'e', 'f'))

print(len(T1)) 4

print(T1[1][0]) 2

print(7 in T1[3]) True

print(T1.index(1)) 0

print(len(T1[3])) 4

print(T1[T1[2][0]]) (7, 'd', 'e', 'f')

Q61. Write the output of the following :

T1 = (1,(2, 'a', (3, 'b', 'c', (7, 'd', 'e', 'f'))))

print(len(T1)) 2

print(T1[1][0]) 2

print(7 in T1[1]) False

print(T1.index(1)) 0

print(len(T1[1])) 3

print(T1[T1[0]]) (2, 'a', (3, 'b', 'c', (7, 'd', 'e', 'f')))

Q62. Write the output of the following :

T1 = (1, (2, 'a') * 2)

print(T1) (1, (2, 'a', 2, 'a'))

print(len(T1)) 2

print(T1[1][1]) a

print(len(T1[1])) 4

Q63. Write the output of the following :

```
T1 = (1, (2, 'a')) * 2
print(T1)
print(len(T1))
print(T1[1][1])
print(len(T1[1]))
print(T1.count(1))
print(T1.index(1))
```

(1, (2, 'a'), 1, (2, 'a'))
4
a
2
2
0

Q64. Write the output of the following :

```
T1 = ('a') * 3
T2 = ('a',) * 3
print(T1)
print(type(T1))
print(len(T1))
print(T2)
print(type(T2))
print(len(T2))
```

aaa
<class 'str'>
3
('a', 'a', 'a')
<class 'tuple'>
3

Q65. Write the output of the following :

```
T1 = "Spider Man:No Way Home"
T2 = (T1.split())
print(T1)
print(type(T1))
print(len(T1))
print(T2)
print(T2[0][0] + T2[1][0])
print(len(T2))
```

Spider Man: No way Home
<class 'str'>
22
['Spider', 'Man:No', 'Way', 'Home']
SM
4

Q66. What type of error is returned by following code :

```
T1 = (1, 2, 3, 4)
T2 = (4, 5, 6)
print(T1 - T2)
```

Type error

Q67. Write the output of the following :

```
T1 = (1, 2, 3, 4)
print(T1 * 3)
print(T1 * (3))
print((T1) * 3)
print(T1 + (3,))
```

(1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4)
(1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4)
(1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4)
(1, 2, 3, 4, 3)

print (tup)

Q68. Write the output of the following :

```
T1 = (10, 20, 30)
print(T1, T1, T1)
print(T1, T1, T1)
print(T1 + T1 + T1)
print(T1 * 3)
```

Q69. Write the output of the following :

```
a = 30
b = (30)
c = (30,)
print(type(a))
print(type(b))
print(type(c))
```

Q70. Write a program to accept a number from the user and create a tuple containing all factors of number. for example :

SAMPLE OUTPUT :

Enter any Number : 9
Tuple is : (1, 3, 9)

Q71. Write a program to arrange all the sub tuples of a tuple in increasing order and store them in another tuple. for example

T1 = ((1, 2, 3), (45, 23), (98, 34, 67)) #Original Tuple

T2 = ((1, 2, 3), (23, 45), (34, 67, 98)) #New Tuple

Q72. Write a program to store the length of all sub tuples of a tuple into a new tuple. for example

T1 = ((1, 2, 3), (45, 23), (98, 34, 67),(34, 56, 78, 9)) #Original Tuple

T2 = (3, 2, 3, 4) # New Tuple

Q73. Write a program to print square of all odd numbers of a tuple.

T1 = (12, 13, 22, 7, 9, 66, 4)

Expected Output : 169, 49, 81

Q74. Write a program to print the common elements of given tuples :

T1 = (1, 2, 3, 4, 5, 6)

T2 = (3, 5, 7, 9, 11)

Expected Output : 3, 5

Write a program to find the union (Common element include once) of given tuples:

T1 = (1, 2, 3, 4, 5, 6)

T2 = (3, 5, 7, 9, 11)

Expected Output : 1, 2, 3, 4, 5, 6, 7, 9, 11

Q76. Write a program to store the sum of all sub tuples of given tuple into a new tuple. for example

T1 = ((1, 2, 3), (45, 23), (98, 34, 67),(34, 56, 78, 9)) #Original Tuple

T2 = (6, 68, 199, 177) # New Tuple

Q77. Write a program to store the maximum value of all sub tuples of given tuple into a new tuple. for example

T1 = ((1, 2, 3), (45, 23), (98, 34, 67),(34, 56, 78, 9),("A", "B", "c"))

#Original Tuple

T2 = (3, 45, 98, 78, c) # New Tuple

Q78. Write a program to display the longest sub tuple from the given tuple.

T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56, 78, 9)) #

Original Tuple

T2 = (34, 56, 78, 9) # New Tuple

Q79. Write a program to display names starting from vowel from the given tuple.

T1 = ("Amit", "Ram", "Esha", "Harry")

Q80. Write a program to print the length of all names in a given tuple.

T1 = ("Amit", "Ram", "Esha", "Harry")

Expected Output : 4, 3, 4, 5

Q81. What is the purpose of following operators in reference to tuples.

a) +

b) *

c) in

print (tup)

Q82. Match the following :

Functions	Description
max( )	Returns the length of the tuple. (5)
min( )	Returns the index value of element in tuple. (4)
count( )	Returns the largest element in tuple. (1)
index( )	Returns the smallest element in tuple. (2)
len( )	Returns the frequency of an element in tuple. (3)

Tuple Questions in Python

Q83. Write the output of the following :

```
>>> S = 1, (2,3,4,5), 51, (61,71)
>>> len(S)  4
>>>S[2 : 3]  51
>>>S[-1][0]  61
>>>type(S)  <class 'tuple'>
```

Q 84. Predict the output:

```
T=(1,2,4,3,8,9)
[T[i] for I in range(0,len(T),2)]
```

[1, 4, 8]

Q 85.What will be the output of the following code snippet?

```
init_tuple=( )
print(init_tuple.__len__())  0
```

1. They are those sequences used to store a tuple of values, of any types. Tuples in python are immutable.

2. T = tuple()

3. T = (2, 4, 6, 8, 10)

4. t = tuple(<1)

5. a = ('q')

14. Lists are mutable in python & lists are better for performing operations while tuples are immutable & they are appropriate for accessing elements.

19. len(): This method returns the length of the tuple
syntax: len(<tuple>)

count(): This method returns the count of a particular element in the tuple
<sequence_name>.count(<object>)

20. ~~tup = eval(input(~~
tup = ()
i = 0
while (i < 5):
    n = int(input("enter number:"))
    tup = tup + (n,)
    i = i + 1
print(tup)

21.
```
F = ()
i = 0
while (i < 5):
    fruit = input ("enter fruit name:")
    F = F + (fruit,)
    i += 1
print (F)
```

22.
```
S = ()
n = int (input (" How many sports do you want: "))
i = 0
while (i < 0):
    sport = ~~s+~~ input ("Enter name:")
    S = S + (sport,)
    i += 1
print (S)
```

32.
```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
t = ()
for i in range (3):
    n1 = int (input (" Enter number: "))
    t = t + (n1,)
T1 = T1 + t
print (T1)
```

36. A tuple inside another tuple is called as a nested tuple.
```
S = ((101, "Amit", 90), (107, "Anishi", 85), (20, "Rashid", 8
```

41.
```
T1 = (12, 17, 18, 25, 19, 12, 18, 5)
n1 = int (input ("enter number:"))
print (T1. count (n1))
```

42.
```
T1 = ("BØ", "O", "O", "K")
st = " ".join (T1)
```

print (st)

43.
```
T1 = (12, 15, 18, 21, 24, 27, 30)
el = int(input(" enter number to remove: "))
if el in T1:
    L = list(T1)
    L.remove(el)
    T1 = tuple(L)
    print(T1)
else:
    print(" element not found")
```

44. Unpacking means creating or initializing individual variables from the values of tuple

```
T1 = (2, 4, 6)
x, y, z = T1
print(x)
print(y)
print(z)
```

45. Creating a tuple from individual values is called packing a tuple.

```
T1 = 2, 4, 6
```

46. Both tuples & strings are immutable.

50.
```
det = ()
tmp = ()
for i in range(3):
    rn = int(input(" enter roll number: "))
    nm = input(" enter name: ")
    sec = input(" enter section: ")
    tmp = (rn, nm, sec)
    det = det + (tmp,)
print(det)
```

51.
```
det = ((1, 'Amita', 'B'), (2, 'Sunita', 'C'), (3, 'Daya',
rn = int(input(" enter roll number :"))
R = 0
for i in range (3):
    if rn == det[i][0]:
        print (det [i])
        R = 1
        break
if R == 0:
    print ("record not found")
```

52.
```
t = ()
for i in range (5):
    n1 = int(input(" Enter number :"))
    t = t + (n1,)
print (" largest number is : ", max(t))
print (" smallest number is : ", min(t))
print (" average is : ", sum(t)/5)
```

53.
```
T1 = (23, 45, 87, 45, 85, 43, 23, 12)
T1 = sorted (T1)
print (T1[-2])
```

54.
```
T1 = (24, 45, 87, 46, 67, 44, 23, 12)
se = 0
so = 0
for i in T1:
    if i%.2 == 0:
        se = se + i
    else:
        so = so+i
print (" sum of even numbers is : ", se)
print (" sum of odd numbers is : ", so)
```

56.
```
n1 = int(input("Enter any number:"))
t = ()
for i in range(1, 11):
    t = t + (n1 * i,)
print(t)
```

57.
```
str = input("Enter string:")
T1 = tuple(str)
print(T1)
```

58.
```
T1 = (1, 2, 3, 'A', 'B', 4, 'sun')
s = 0
for i in T1:
    if str(i).isdigit():
        s = s + i
print(s)
```

59.
```
T1 = (("Suman", 75), ("Gloary", 35), ("Ravi", 50))
s = 0
for i in i[1]:
    s = s + i[1]
print("Sum = ", s)
```

70.
```
n1 = int(input("enter number:"))
t = ()
for i in range(1, n1+1):
    if n1 % i == 0:
        t = t + (i,)
print(t)
```

71.
```
T1 = ((1, 2, 3), (45, 23), (98, 34, 67))
T2 = ()
for i in T1:
    T2 = T2 + (tuple(sorted(i)),)
print(T2)
```

72.
```
T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56,
T2 = ()
for i in T1:
    T2 = T2 + (len(i),)
print (T2)
```

73.
```
T1 = (12, 13, 22, 7, 9, 66, 4)
for i in T1:
    if i%2!=0:
        print (i**2, end = " ")
```

74.
```
T1 = (1, 2, 3, 4, 5, 6)
T2 = (3, 5, 7, 9, 11)
for i in T1:
    if i in T2:
        print (i, end = ' ')
```

75.
```
T1 = (1, 2, 3, 4, 5, 6)
T2 = (3, 5, 7, 9, 11)
T3 = T1
for i in T2:
    if i not in T1:
        T1 = T1 + (i,)
print (T1)
```

76.
```
T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56, 78,
T2 = ()
for i in T1:
    T2 = T2 + (sum(i),)
print (T2)
```

77.
```
T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56, 78, 9)
T2 = ()
for i in T1:
    T2 = T2 + (max(i),)
```

print (T2)

18. T1 = ((1,2,3), (45, 23), (98, 34, 51) (34, 56, 78, 9), (A, B,
T2 = ()
for i in T1:
    T2 = T2 + (len(i),)
ind = T2.index (max (T2))
print (T1[ind])

19. T1 = ('Amit', 'Ram', 'Esha', 'Mary')
for i in T1:
    if i[0] in ('AEIOUaeiou'):
        print (i)

20. T1 = ('Amit', 'Ram', 'Esha', 'Mary')
for i in T1:
    print (len (i), end = ' ')

21. (a) The (+) operator is used to join two tuples
>>> (1, 2, 3) + (5, 6, 7)
>>> (1, 2, 3, 5, 6, 7)

(b) The (*) operator repeats the elements in the tuple
>>> (1, 2, 3) * 3
>>> (1, 2, 3, 1, 2, 3, 1, 2, 3)

(c) The (in) operator checks whether an element is present inside a tuple.
>>> 1 in (1, 2, 3)
>>> True